Problem Set 3

Starter

1 (E)**.** Let $V = \left\{ \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} : a, b \in \{-1, 1\} \right\}$. Show that V is not cyclic.

2 (E)**.** Show that $(\mathbb{R}, +)$ is generated by the squares of numbers of $\mathbb{R}$.

3 (E)**.** Consider the additive group $(\mathbb{Z}, +)$. Find all integers m such that $\langle m \rangle = \langle 3, 4, 6 \rangle$, and find all integers n such that $\langle n \rangle = \langle 3 \rangle \cap \langle 4 \rangle \cap \langle 6 \rangle$.

4 (L-M)**.** Let $C = \langle x \rangle$ be a cyclic group of order n. Show that the order of x^k is $\frac{n}{\gcd(k,n)}$ for every integer k. Deduce that $\langle x \rangle = \langle x^k \rangle$ if and only if $\gcd(k, n) = 1$.

Advanced

5 (L-M)**.** Show that a group with no nontrivial proper subgroups must be cyclic of order 1 or a prime.

6 (M)**.** Show that every cyclic group is abelian. Show that not every abelian group is cyclic.

Recall that the proper subgroups of a group G are the subgroups of G not equal to G. If every proper subgroup of a group G is cyclic then must G be abelian? Justify your answer.

7 (M)**.** Let G be a finite group of order n. Suppose that G is a cyclic group. Prove that G has a unique subgroup of order d for each positive divisor d of n.

8 (H)**.** Let G be a finite group of order n. Suppose that G has a unique subgroup of order d for each positive divisor d of n. Prove that G is a cyclic group.